

Design and Implementation of an Isolated 10W Dual-Output Flyback Converter

Submitted by:

Ojas Rajesh Thete (230108034)
Vancha Avinash Reddy (230108059)

Supervisor:

Dr. Nawaz Hussain

Evaluators:

Dr. Shabari Nath
Dr. A. Rajesh
Dr. Praveen Tripathy

EE396 Design Lab

Department of Electrical and Electronics Engineering
Indian Institute of Technology Guwahati

April 2026

Design and Implementation of an Isolated 10W Dual-Output Flyback Converter

Ojas Rajesh Thete (230108034), Vancha Avinash Reddy (230108059)

EE396 Design Lab, IIT Guwahati

1 Circuit Diagram and Implementation

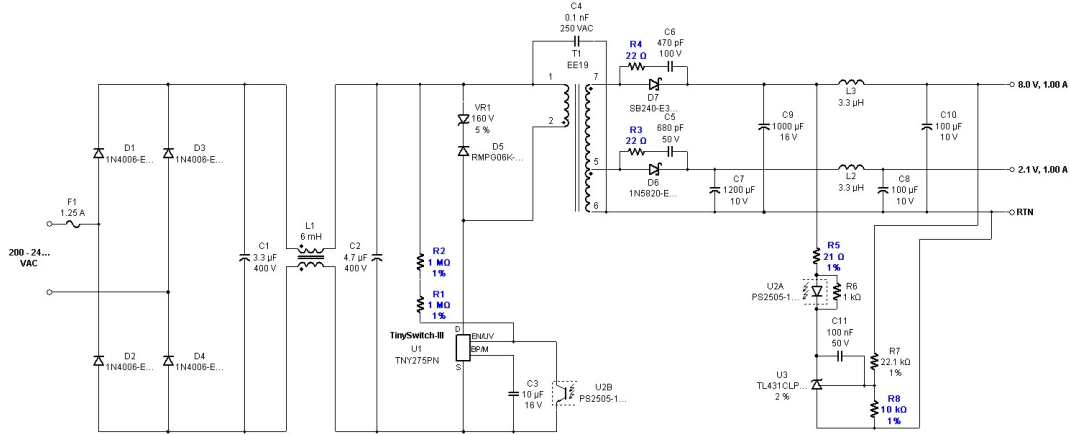


Figure 1: Schematic for the 8.0 V / 2.1 V Flyback Converter

The circuit is designed as an isolated flyback converter providing two distinct voltage rails. The primary side utilizes a TinySwitch-III TNY275PN controller for high-frequency switching and integrated protection.

Table 1: Component Identification and Specifications

Label	Component	Value/Model	Function
D1-D4	Bridge Rectifier	1N4007-E	Input Rectification
U1	Switch IC	TNY275PN	PWM Controller/MOSFET
T1	Transformer	EE25 Ferrite	Energy Storage/Isolation
VR1	TVS Diode	160 V, 5%	Primary Drain Clamp
U2	Optocoupler	PS2505-1	Feedback Isolation
U3	Shunt Regulator	TL431	Error Amplifier
R1, R2	Resistors	1 MΩ, 1%	Undervoltage Detection
R7, R8	Resistors	22.1 kΩ, 10 kΩ	Output Voltage Divider
L3, L2	Chokes	3.3 μH	Output Post-Filter
C9, C7	Bulk Caps	1000 μF, 1200 μF	Secondary Storage

2 PCB Layout and Implementation

The circuit was implemented on a Zero PCB after initial layout design. To minimize EMI and parasitic inductance, high-current paths on the secondary side were kept as short as possible. The following figures illustrate the hardware implementation and the proposed PCB design in a single vertical column.

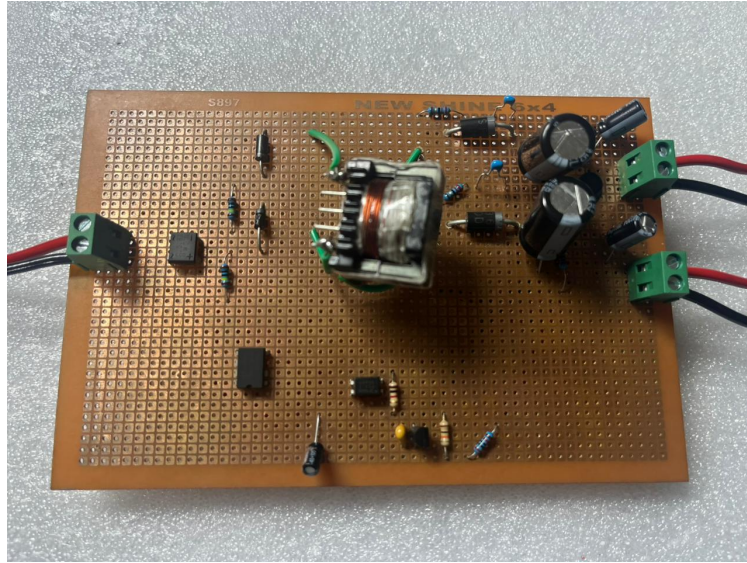


Figure 2: Hardware Implementation on Zero PCB

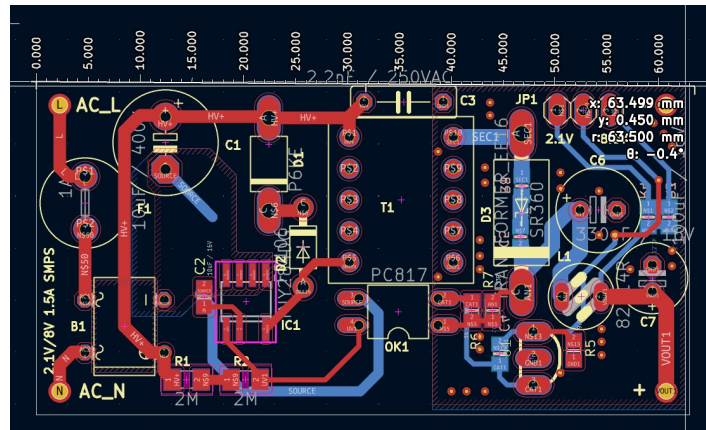


Figure 3: Proposed PCB Trace Layout

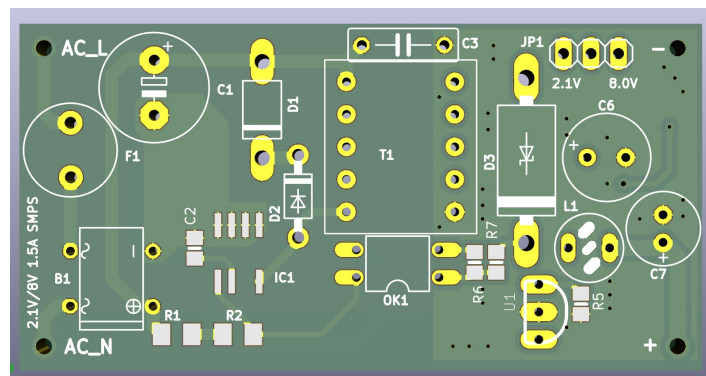


Figure 4: 3D Front View of the Custom PCB Design

3 Description

The flyback SMPS converts a 220 V AC input into regulated DC outputs. The input AC is rectified by a bridge and smoothed by a Pi-filter (C1, L1, C2). The TNY275PN drives the transformer primary, storing energy in the core during the ON-state. Upon the switch turning OFF, the energy is transferred to the secondary windings.

In this design, the secondary side utilizes a stacked winding topology to improve cross-regulation and efficiency. Rather than using separate independent windings, the 2.1 V rail is obtained as a specific tap-out from the 8.0 V secondary winding. The 8.0 V rail is sensed via the TL431, which drives the optocoupler to modulate the primary-side duty cycle, ensuring stable regulation for both output rails.

4 Calculations

4.1 AC Input to DC Conversion

Given $V_{AC} = 220$ V (rms):

$$V_m = \sqrt{2} \cdot 220 = 311.1 \text{ V} \quad \text{and} \quad V_{IN} = \frac{2V_m}{\pi} \approx 198.1 \text{ V} \quad (1)$$

4.2 Output Requirement

Total power for the dual-rail system:

$$P_O = (8V \times 1A) + (2.1V \times 1A) = 10.1 \text{ W} \quad (2)$$

4.3 Input Power and Loss

Assuming an efficiency (η) of 0.74:

$$P_{IN} = \frac{P_O}{\eta} \approx 13.65 \text{ W}; \quad P_{LOSS} = P_{IN} - P_O = 3.55 \text{ W} \quad (3)$$

4.4 Duty Ratio (D)

Determined by primary inductance $L_P = 1300$ μ H and secondary $L_S \approx 2$ μ H:

$$n = \frac{N_P}{N_S} = \sqrt{\frac{1300}{2}} \approx 25.5 \implies D \approx 0.528 \text{ (52.8\%)} \quad (4)$$

4.5 Transformer Primary Inductance (L_P)

Calculated for $f_s = 132$ kHz:

$$E = \frac{P_{IN}}{f_s} \approx 1.03 \times 10^{-4} \text{ J}; \quad I_{P,pk} = \sqrt{\frac{2E}{L_P}} \approx 0.40 \text{ A} \quad (5)$$

Verified measured value: $L_P = 1300$ μ H.

4.6 Capacitor Ripple Current Verification

RMS ripple current calculation for thermal stability:

$$I_{C,rms} = \sqrt{I_{S,rms}^2 - I_O^2} \approx \sqrt{4.0^2 - 1.0^2} \approx 3.87 \text{ A} \quad (6)$$

4.7 Feedback Network Verification

With $R7 = 22.1 \text{ k}\Omega$ and $R8 = 10 \text{ k}\Omega$:

$$V_O = 2.495 \cdot \left(1 + \frac{22.1}{10}\right) = 8.01 \text{ V} \quad (7)$$

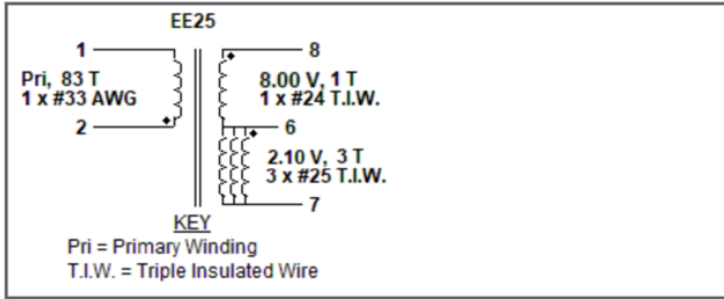
4.8 Output LC Filter Calculation

Cutoff frequency for $L_3 = 3.3 \text{ }\mu\text{H}$ and $C_{OUT} = 1000 \text{ }\mu\text{F}$:

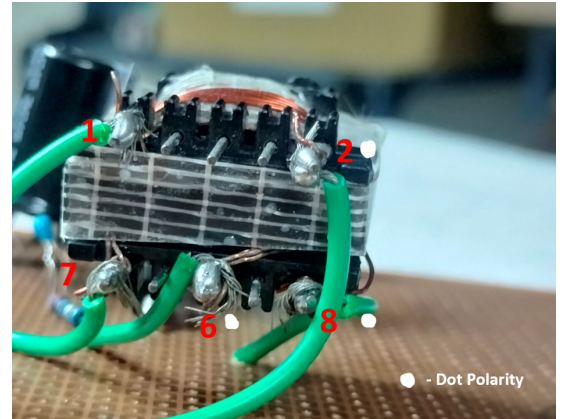
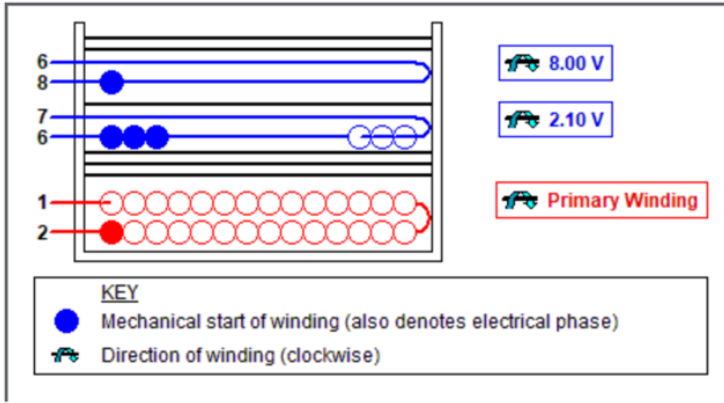
$$f_c = \frac{1}{2\pi\sqrt{3.3 \cdot 10^{-6} \cdot 1000 \cdot 10^{-6}}} \approx 2.77 \text{ kHz} \quad (8)$$

5 Transformer Construction

▼ Electrical Diagram



▼ Mechanical Diagram



(a) Electrical, Mechanical, and Winding Diagrams

(b) Physical Pinout and Winding Starts

Figure 5: Transformer (T1) Design Specifications and Physical Implementation

The transformer was wound on an **EE25** core gapped for 197 nH/T^2 with a **0.5 mm air gap**.

- **Primary:** 83 turns of 33 AWG wire (Pin 2 to 1).
- **Secondary (2.10V):** 3 turns of #25 T.I.W. (Pin 6 to 7) derived as a tap.
- **Secondary (8.00V):** 1 turn of #24 T.I.W. (Pin 8 to 6).
- **Finish:** Polyester insulation tape and uniform dip varnishing.



(a) L_P : 1.335 mH



(b) $L_{2.1V}$: 2.17 μH



(c) $L_{8.0V}$: 1.21 μH

Figure 6: LCR Meter Verification of Transformer Inductances

6 π -Filter Implementation challenge

When a π -filter was introduced at the input, the circuit failed during startup due to excessive inrush current charging the input capacitors. At power-on, the capacitors initially behave as short circuits; if the series inductor parasitic resistance is low, it cannot effectively limit the transient current spike. This transient caused failure in the feedback optocoupler, which serves as the sensitive interface between high-voltage primary and low-voltage secondary.

7 Experimental Results



(a) 8.0V Rail ($V_{max} \approx 7.89\text{V}$)



(b) 2.1V Rail ($V_{max} \approx 2.37\text{V}$)

Figure 7: Steady-State DSO Output Waveforms

8 Conclusion

Experimental hardware confirms regulated 8.0 V and 2.1 V rails aligning with specifications. Measurements ($L_P = 1.335 \text{ mH}$) verify construction integrity. Future work will focus on implementation of the PCB design on a green PCB and improved EMI performance by incorporating a π -filter. Such designs are widely used in low-power electronic systems, embedded devices and battery chargers due to their ability to provide a regulated DC output efficiently.